

EliGSiR: Continual RGB-D Mapping with Gaussian Splatting under Bounded Compute

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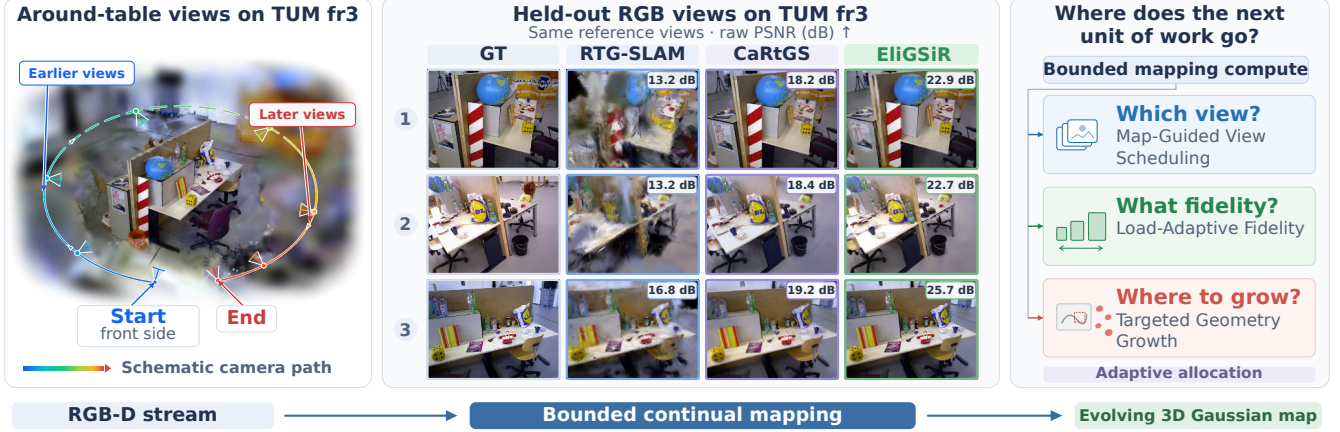


Fig. 1: **From a continual RGB-D stream to an evolving 3D Gaussian map.** EliGSiR allocates bounded compute across view scheduling, supervision fidelity, and geometry growth, yielding higher-quality held-out renderings than RTG-SLAM and CaRtGS in the shown views.

Abstract: Conventional 3D Gaussian Splatting assumes a closed set of observations and long optimization schedules. Continual RGB-D mapping in contrast poses the problem that new observations arrive online, while previously reconstructed regions must be preserved. We present EliGSiR¹ (Evidence-guided Load-adaptive Incremental Gaussian Splatting with Image Replay), a continual Gaussian mapper that controls how the available optimization budget is used as the reconstruction evolves. Map-Guided View Scheduling filters redundant incoming views and reconsiders retained views according to the current state of the map. Load-Adaptive Fidelity adjusts supervision resolution to the current mapping load instead of following a fixed resolution schedule. Targeted Geometry Growth separates depth supervision from Gaussian creation and adds geometric capacity only where repeated RGB-D observations indicate missing or misplaced structure. Together, these mechanisms adapt which views are optimized, how much image detail is used, and where the representation grows while mapping remains active. We evaluate EliGSiR on Replica, TUM RGB-D, ScanNet++, and real RGB-D sensor sequences, considering both the final reconstruction and the map available throughout acquisition. On TUM RGB-D fr3/long_office_household, EliGSiR reaches 21.52 dB with the same ground-truth mapping poses used by the controlled baselines, compared with 19.42 dB for SplatAM. In the tracked-pose comparison, EliGSiR with live ORB-SLAM3 poses reaches 23.02 dB in 155.5s, compared with 20.10 dB in 230.9s for CaRtGS using its native tracker. We further evaluate reconstruction throughout acquisition and show how EliGSiR’s adaptive view scheduling, supervision fidelity, and geometry growth improve the use of the available mapping budget.

I. INTRODUCTION

Robots operating in real environments require dense 3D representations while they are moving. Such maps support perception, inspection, navigation, and manipulation, but they must already be useful before the full environment has been observed. This differs from offline reconstruction, where all observations are available in advance and optimization can continue until convergence. A continual mapper must instead extend its representation as new parts of the scene become visible while preserving previously reconstructed content.

3D Gaussian Splatting (3DGS) [1] is especially attractive for this setting because it combines an explicit scene representation with fast differentiable rendering. Recent systems have adopted Gaussian Splatting for RGB-D SLAM and online mapping [2]–[5]. These methods demonstrate that Gaussian maps can be built online and, in some cases, in real time. Their optimization must handle a continuous sensor stream introducing new observations.

This creates a resource-allocation problem inside the mapping process. Compute spent on a redundant view is unavailable for newly observed or under-optimized regions. Expensive high-resolution updates can reduce the number of optimization steps that fit into the available time, while aggressive map growth increases the representation that must subsequently be optimized. The mapper must therefore balance revisiting useful observations, choosing an appropriate supervision fidelity, and extending the representation only where additional geometric support is needed.

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We present EliGSiR, a continual RGB-D Gaussian mapper designed around these three runtime decisions (Fig. 1). EliGSiR continuously adapts where its optimization budget is spent as both the reconstruction and the current workload change. Instead of committing these decisions once when an observation arrives, the mapper can revisit earlier choices and redirect computation toward parts of the reconstruction that currently benefit from further optimization.

Our main contributions are:

- **Map-Guided View Scheduling** combines inexpensive redundancy filtering with continued reprioritization of retained views according to the evolving reconstruction.
- **Load-Adaptive Fidelity** makes supervision resolution a reversible runtime decision driven by the current mapping load rather than a predefined resolution schedule.
- **Targeted Geometry Growth** decouples depth supervision from Gaussian creation and introduces additional capacity where repeated RGB-D observations indicate missing or misplaced geometry.

II. RELATED WORK

Online Gaussian SLAM and mapping. 3D Gaussian Splatting has been adopted by several systems for online reconstruction and SLAM. SplatAM [2] jointly tracks the camera and reconstructs an RGB-D scene using silhouette-guided map expansion and keyframe-based optimization. MonoGS [3] couples tracking and mapping through differentiable Gaussian rendering, while Gaussian-SLAM [6], Photo-SLAM [7], GS-SLAM [8], RTG-SLAM [4], and GS-ICP-SLAM [5] combine Gaussian reconstruction with different tracking and map-update strategies. LoopSplat [9] additionally addresses global consistency through Gaussian submap registration. They demonstrate the feasibility of online Gaussian map reconstruction, but differ in how compute resources are allocated and how views are scheduled for optimization.

EliGSiR focuses specifically on how this mapping budget can be redistributed as the reconstruction changes.

View selection and replay. Selecting a limited set of observations for repeated optimization is common in dense neural mapping. NICE-SLAM [10] and Point-SLAM [11] use keyframes to revisit earlier observations while incorporating new input. Online Gaussian mappers retain selected historical views and use recency, motion, overlap, or covisibility to control which observations participate in optimization. This avoids redundant work, but neglects that earlier views can become useful again as the reconstruction evolves. **Map-Guided View Scheduling** therefore keeps retained views available for continued reprioritization instead of treating their initial admission decision as permanent. Spatial reconstruction error further allows replay to be directed toward parts of the map that currently require additional optimization.

Compute-aware optimization and supervision. Real-time Gaussian mapping has motivated several strategies for reducing or redistributing optimization cost. RTG-SLAM [4] separates stable and unstable Gaussian populations and restricts optimization and rendering to the unstable part of

the representation. CaRtGS [12] addresses computational imbalance by adapting training effort and densification during online Gaussian optimization. Photo-SLAM [7] uses Gaussian-pyramid training to progress from coarse to finer image levels as optimization proceeds. **Load-Adaptive Fidelity** instead treats supervision resolution as a reversible runtime variable: image fidelity follows the current mapping load and can dynamically adjust between coarse and fine levels throughout acquisition rather than following a predefined progression.

Geometry growth from RGB-D observations. Depth provides direct evidence for initializing and extending Gaussian maps. SplatAM [2] adds Gaussians according to silhouette and depth evidence, while RTG-SLAM [4] introduces primitives in newly observed regions and where color or depth errors remain large. GS-SLAM [8] also adapts Gaussian expansion during mapping, and GSFusion [13] combines a TSDF representation with image-space subdivision to obtain compact Gaussian geometry. 3DGS-MCMC [14] provides a mechanism for relocating, replacing, and densifying Gaussian primitives. **Targeted Geometry Growth** differs in the decision that triggers map expansion: depth can first supervise the existing representation without requiring new primitives, while additional capacity is introduced only where repeated RGB-D observations support missing or misplaced geometry.

Online optimization under bounded compute. The related methods reduce online reconstruction cost through different combinations of keyframe selection, selective updates, multi-resolution training, controlled densification, and compact representations. These strategies establish that real-time Gaussian mapping is feasible, but address different parts of the computational problem. EliGSiR treats the available mapping budget as a runtime constraint shared between selecting retained views for optimization, the dynamic resolution for supervision, and the region for optimizing Gaussians.

III. PROBLEM FORMULATION

We consider continual RGB-D mapping of a static scene with known camera poses. At time i , the mapper receives an RGB image I_i , a depth map D_i , camera intrinsics K_i , pose $T_{wc,i}$, and timestamp τ_i , constituting an observation $o_i = (I_i, D_i, K_i, T_{wc,i}, \tau_i)$. Only arrived observations can update the current Gaussian map Θ_i . Camera poses may be provided by a reference trajectory or a tracking frontend.

The observation stream and map optimization proceed at different rates. Under bounded compute, the mapper cannot assume that every retained view can be optimized equally and at full resolution. Newly observed depth does not necessarily justify immediate growth of the Gaussian representation. The available optimization budget must therefore be distributed across the runtime decisions about frame scheduling, image fidelity, and map growth.

Earlier observations may be retained for later optimization, to which we refer as *replay*. Their usefulness can change as the reconstruction evolves: a view may become less important once its region is well reconstructed, or more useful again when later updates reveal remaining errors or degrade previously reconstructed content.

Our objective is to keep the map responsive to the incoming stream while directing the available compute toward the work that currently benefits the reconstruction quality most.

IV. METHOD

EliGSiR keeps the Gaussian representation available for standard photometric and depth-based optimization, but changes how the available mapping budget is allocated while the RGB-D stream is active. It makes three runtime decisions (Figure 2): which views receive optimization, at what image fidelity they are supervised, and where additional geometric capacity is introduced. Map-Guided View Scheduling determines which current and retained views are used for optimization, Load-Adaptive Fidelity controls the cost of their supervision, and Targeted Geometry Growth adds map capacity where the current representation is insufficient.

A. Map-Guided View Scheduling

The utility of an observation is not fixed when it enters the map. An admitted view may be important while its region is first reconstructed, lose priority once that region is well explained, and become useful again if later optimization exposes remaining errors. EliGSiR therefore separates the initial admission of a view from its longer-term priority for supervision.

Initial filtering and admission. Incoming observations first pass a lightweight admission stage. A pose-based pre-filter removes redundant views before map-based evaluation. The remaining candidates are compared with the current Gaussian map, using view overlap to identify observations that provide additional coverage. If no suitable view has been admitted for a fixed interval, the next suitable candidate is retained through a maximum-gap rule. This keeps the admission stage inexpensive while avoiding a fixed-stride keyframe policy.

Continuous supervision-set refinement. Admission does not assign a permanent keyframe status. EliGSiR maintains an active supervision set together with lower-priority retained views. The active set contains observations currently selected for repeated optimization, while retained views remain available for later replay.

These priorities are updated as the reconstruction changes. A retained view can return to active supervision when it becomes more useful, and can lose priority again once the region it covers is sufficiently reconstructed. Historical observations therefore remain available for later optimization instead of being discarded after their initial ranking.

Spatial supervision focus. View-level scheduling alone can still spend unnecessary compute on already well reconstructed regions. EliGSiR therefore divides the observed scene into fixed world-space microcells and accumulates reconstruction error from training renders that are already produced during optimization. The resulting spatial signal indicates which parts of the map remain weak or have regressed.

Views covering these regions receive increased replay priority, while views dominated by well-reconstructed regions are selected less often. This allows supervision to shift toward parts of the scene that currently need additional optimization without losing access to the retained observation history.

B. Load-Adaptive Fidelity

Selecting a useful view does not determine how much compute should be spent on its optimization. Full-resolution supervision preserves the most image detail, but also makes each update more expensive. When newly observed scene content increases the amount of mapping work, performing every update at full resolution can reduce the number of optimization steps that fit into the available time. EliGSiR therefore treats supervision fidelity as a runtime decision.

The scheduler measures the recent mapping cost and selects from a small set of discrete image resolutions, prioritising high resolutions when sufficient compute is available, and reduces resolution as mapping load increases.

The decision is reversible: supervision can return to higher resolution when the mapping load drops and decrease again when the workload rises. Supervision fidelity therefore follows the current mapping load rather than a predefined coarse-to-fine schedule.

Changing the supervision resolution does not modify the retained source observation. The original RGB-D data remains available, and the camera intrinsics are scaled consistently with the image grid. EliGSiR can therefore reduce supervision cost temporarily without discarding the image detail available for later optimization.

C. Targeted Geometry Growth

Depth supervision and Gaussian creation serve different purposes. A depth observation can constrain Gaussians that already represent a surface without requiring new primitives. Newly observed or insufficiently represented regions, in contrast, require additional geometric support. EliGSiR therefore separates the use of depth for supervision from the decision to grow the representation.

Sparse structural support. Newly observed space first receives a bounded set of sparse depth-derived Gaussians. Rather than densely converting all depth into map geometry, these initial primitives establish the broad structure required to begin reconstruction. A single incoming observation can therefore extend the map without immediately introducing a dense Gaussian population. This initial support is not treated as the final geometry. Retained RGB-D observations remain available after their first use, allowing the representation to be refined as further observations become available.

Progressive geometry growth. When a region is revisited, EliGSiR compares the observed surface with the support already provided by the Gaussian map. Additional insertion is suppressed where the current representation adequately explains the observations and directed toward regions that remain unsupported or geometrically misplaced. Repeated observations therefore provide additional evidence before finer geometry is introduced, rather than allowing each unsupported depth measurement to trigger immediate map growth.

Depth consequently has two roles. It supervises geometry that already exists and provides evidence for extending regions that remain insufficiently represented. The first occurs during ordinary optimization; the second changes the capacity of the representation and is controlled separately.

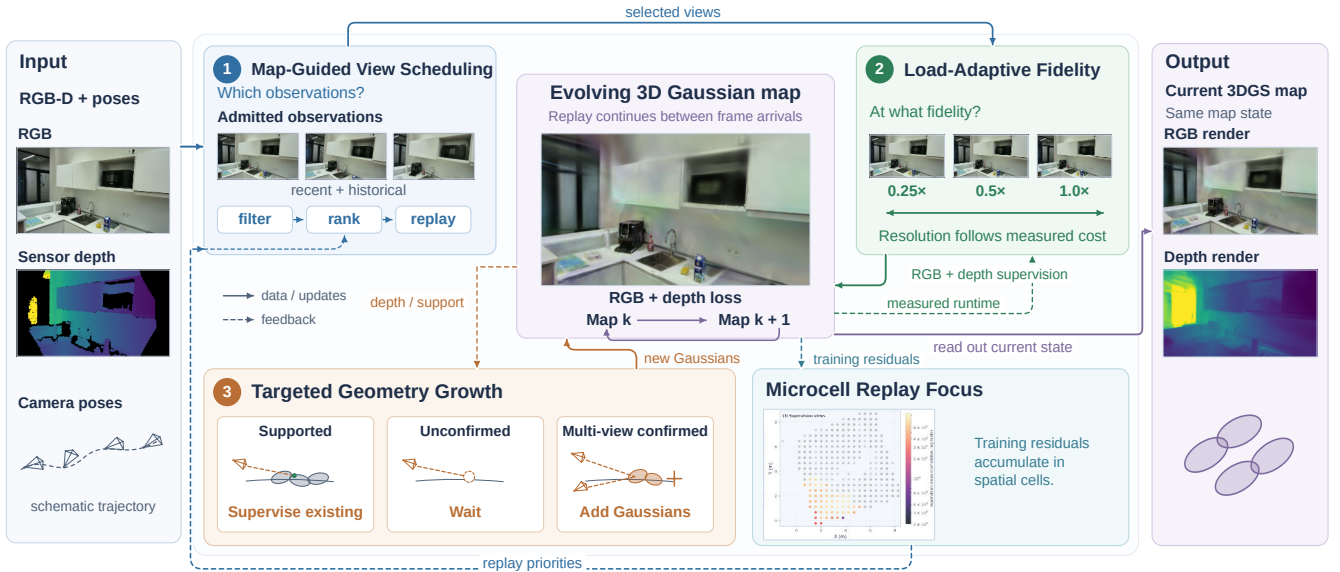


Fig. 2: **Method overview.** EliGSiR allocates mapping work across observation selection and replay, supervision fidelity, and geometry growth. Retained observations provide recent and historical supervision, measured mapping cost controls image resolution, and RGB-D evidence guides where the Gaussian representation grows.

Population refinement. We use the energy-based 3DGS-MCMC strategy [14] for Gaussian relocation, replacement, and densification. MCMC provides the population-update mechanism rather than constituting a separate contribution of EliGSiR. Together with sparse initial support and later targeted insertion, it allows the Gaussian population to refine as additional observations and optimization become available.

The same Gaussian representation remains available after the input stream ends. If additional refinement is performed, optimization can continue at higher fidelity and further refine the Gaussian population. The online mechanisms therefore reduce unnecessary work during acquisition without replacing the map with a separate reduced representation.

V. EVALUATION

A. Experimental Setup

Following common evaluation settings in recent dense and Gaussian SLAM work [2], [4], we evaluate on representative scenes from Replica, TUM RGB-D, and ScanNet++. Specifically, we use Replica *room2* [15], TUM RGB-D *fr3/long_office_household* and *fr1/desk* [16], ScanNet [17] and ScanNet++ *8b5caf3398* [18]. Additional cross-scene evaluation uses Replica *office0* and the real-sensor Orbbec sequences *floor2* and *kitchen1*. Qualitative cross-dataset evaluation additionally includes ScanNet. These scenes cover synthetic reconstruction, real handheld RGB-D acquisition, and high-quality novel-view evaluation while overlapping with sequences used by the evaluated baselines. Frames are processed in temporal order, while admitted observations may remain available for later replay. Optimization can therefore revisit previously observed views, but never access observations that have not yet arrived.

The main Replica experiments run on an AMD Radeon R9700 GPU, while the main TUM experiments use an

NVIDIA RTX 4070. EliGSiR is implemented in PyTorch and *gsplat* [19], with CUDA and ROCm support for NVIDIA and AMD GPUs. The same implementation also runs on integrated platforms such as AMD Strix Halo and is available as a ROS 2 package. Camera poses are supplied through a generic interface and can originate from ground truth, visual odometry, or a SLAM frontend. Baselines run with their original implementations and native rasterizers. For cross-method evaluation, exported Gaussian maps are additionally rendered through a common *gsplat* path.

B. Evaluation Protocol

Held-Out Views. A fixed subset of frames is excluded from map optimization and used for evaluation. Within each comparison set, all methods use the same held-out split. During mapping, these views are rendered every 100 optimization updates.

Rendering Consistency. All held-out evaluations use identical camera poses, intrinsics, and reference images. Exported maps are checked with both the native rasterizer, where available, and a common *gsplat* path to ensure that renderer differences do not affect the comparison.

Pose Sources. Evaluation poses are fixed, while mapping poses follow the method setup. All controlled comparisons use the same RGB-D stream and ground-truth poses. CaRtGS is the only exception because it does not support externally supplied poses and therefore uses its native tracking trajectory. For comparison, we also run EliGSiR with live ORB-SLAM3 poses [20].

Map State. We report the evaluated map state because methods differ in how much processing continues after the input stream. ME denotes EliGSiR after the stream and bounded drain, NC the completed native pipeline of RTG-SLAM or SplaTAM, and NT the short native post-stream

tail of CaRtGS. $+Nk$ PM denotes N thousand additional post-mapping updates; VarSplat’s $+30k$ PM includes global optimization and bundle adjustment.

Mapping-End Quality. Mapping-end quality is measured after the input RGB-D sequence and a short drain of already scheduled work. Evaluation rendering is excluded from the reported mapping time. We report reconstruction metrics together with elapsed mapping time, optimizer updates, supervised views, dropped frames, and Gaussian count where applicable.

For TUM RGB-D, we additionally optimize an offline reference using all available frames and ground-truth poses to estimate the reconstruction quality attainable without the online compute constraint. This result is reported separately and is not treated as an online baseline.

Online Mapping Quality. Mapping-end metrics do not capture reconstruction quality during acquisition. We therefore report held-out PSNR over both elapsed wall-clock time and cumulative supervised views, separating runtime progress from supervision efficiency. Temporal partitions further reveal whether parts of the sequence fall behind during mapping.

We also report supervised pixels, obtained by summing the image resolution of every supervised view over all optimization updates. Values are given in megapixels (MPix) and quantify the total amount of image-space supervision.

Regional quality. Global image metrics can hide local reconstruction failures. We therefore evaluate fixed 0.5 m world-space cells. Each cell c has a fixed set Ω_c of held-out reference samples defined by camera, pixel, and reference depth.

Eligibility is independent of the mapper. A cell becomes eligible at $t_{\text{vis}}(c)$, the first evaluation checkpoint after its surface has been observed from at least two distinct source views; held-out frames do not count. It is measurable when at least 300 reference samples from at least two held-out views are available. We denote cells that are both eligible and measurable by $\mathcal{M}(t) = \{c \mid c \text{ is eligible and measurable at } t\}$. All regional quality metrics below are computed over $\mathcal{M}(t)$.

For each measurable cell, we compute RGB mean squared error

$$e_c(t) = \frac{1}{3|\Omega_c|} \sum_{(v,u) \in \Omega_c} \sum_{k \in \{R,G,B\}} \left(\hat{I}_{t,v}^k(u) - I_v^{\text{ref},k}(u) \right)^2,$$

and convert it to cell-level PSNR,

$$q_c(t) = -10 \log_{10}(\max\{e_c(t), \varepsilon\}), \quad \varepsilon = 10^{-12},$$

with RGB values normalized to $[0, 1]$.

We parameterize acquisition progress by

$$p(t) = \frac{N_{\text{released}}(t)}{N_{\text{total}}},$$

where $N_{\text{released}}(t)$ is the number of source frames released by checkpoint t .

Causal Visible Quality (CVQ) is the mean cell-level PSNR over the currently evaluable scene:

$$\text{CVQ}(t) = \frac{1}{|\mathcal{M}(t)|} \sum_{c \in \mathcal{M}(t)} q_c(t).$$

Causal Usable Coverage (CUC@Q) is the fraction of evaluable cells above a quality threshold Q :

$$\text{CUC@Q}(t) = \frac{|\{c \in \mathcal{M}(t) \mid q_c(t) \geq Q\}|}{|\mathcal{M}(t)|}.$$

We use $Q = 20$ dB.

To summarize reconstruction quality throughout acquisition, we integrate CVQ over normalized source progress:

$$\text{CVQ-AUC} = \frac{\int_{p_{\min}}^{p_{\max}} \text{CVQ}(p) dp}{p_{\max} - p_{\min}}.$$

The cell partition, visibility criteria, and quality threshold are fixed for all compared methods; missing renders count as failures.

VI. RESULTS

We first compare final-map reconstruction across benchmark and real-sensor sequences. We then evaluate reconstruction quality during acquisition and continued refinement, isolate the effects of the three runtime decisions, and finally provide qualitative comparisons.

The final-map comparison, causal ablations, and refinement study use separate runs and evaluation splits; absolute PSNR values should therefore not be compared across these experiment blocks.

A. Final-Map Reconstruction

Table I summarizes final-map reconstruction on the controlled TUM RGB-D comparison and additional benchmark and real-sensor sequences.

On TUM RGB-D `fr3/long_office_household`, the controlled comparison separates common ground-truth mapping poses from tracked poses. With the same ground-truth poses, EliGSiR reaches 21.52 dB, compared with 19.42 dB for SplatAM, 16.44 dB for VarSplat, and 14.46 dB for RTG-SLAM. With live tracking, EliGSiR reaches 23.02 dB, compared with 20.10 dB for CaRtGS using its native tracker.

Figure 3 shows the trade-off between reconstruction quality and processing time. With live tracking, EliGSiR reaches 23.02 dB at a real-time factor of $1.78\times$, compared with 20.10 dB at $2.65\times$ for CaRtGS. The GT-pose baselines require $13.17\text{--}28.64\times$ the sequence duration to reach their reported final maps. Their final PSNR should therefore be read together with the processing time needed to obtain it.

Across the additional scenes, EliGSiR reaches 23.50 dB on `kitchen1`, 37.07 dB on `Replica room2`, and 19.58 dB on `ScanNet++ 8b5caf3398`.

B. Quality During Mapping and Continued Refinement

We next evaluate reconstruction quality throughout acquisition. Figure 4 compares the full method and its ablations over mapping time.

The full method reaches a CVQ-AUC of 21.18 dB, compared with 20.62 dB without Load-Adaptive Fidelity and 20.84 dB without Targeted Geometry Growth. Although fixed native-resolution supervision reaches a higher final global PSNR, the full method achieves a higher CVQ-AUC while

TABLE I: **Reconstruction quality and processing time across RGB-D scenes.** Methods are reported at their evaluated map states using their native finalization procedures. Final-map quality and elapsed mapping time should be considered jointly, since some native pipelines continue substantial optimization after acquisition. Reported times are descriptive because hardware and timing scopes differ across experiments.

Method	Map state	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Time [s]
<i>TUM RGB-D fr3/long_office_household: GT mapping poses</i>					
SplaTAM	NC	19.42	0.703	0.445	1382.6
RTG-SLAM	NC	14.46	0.553	0.614	1147.8
VarSplat	+30k PM	16.44	0.574	0.510	2495.5
EliGSiR	ME	21.52	0.757	0.320	176.1
<i>TUM RGB-D fr3/long_office_household: tracked poses</i>					
CaRtGS	NT	20.10	0.689	0.363	230.9
EliGSiR	ME	23.02	0.803	0.276	155.5
<i>Replica office0</i>					
RTG-SLAM	NC	34.62	0.940	0.263	412.4
VarSplat	+30k PM	40.91	0.972	0.203	3742.2
EliGSiR	ME	34.20	0.938	0.223	324.0^a
<i>Replica room2</i>					
EliGSiR	+2k PM	37.07	0.964	0.137	746.4
<i>Orbbec floor2</i>					
RTG-SLAM	NC	18.52	0.718	0.461	775.2
SplaTAM	NC	21.40	0.782	0.390	18349.6
VarSplat	+30k PM	25.88	0.883	0.254	2274.7
EliGSiR	ME	16.27	0.601	0.510	555.1
<i>Orbbec kitchen1</i>					
RTG-SLAM	NC	16.75	0.669	0.506	408.0
SplaTAM	NC	17.59	0.731	0.447	2456.7
CaRtGS	NT	20.12	0.856	0.281	99.2^b
EliGSiR	ME	23.50	0.840	0.248	203.3
<i>ScanNet++ 8b5caf3398</i>					
EliGSiR	ME	19.58	0.817	0.307	390.9
<i>TUM RGB-D fr1/desk</i>					
RTG-SLAM	NC	16.07	0.559	0.570	707.0
SplaTAM	NC	18.75	0.659	0.471	4726.5
EliGSiR	+30k PM	20.60	0.729	0.408	678.3

^a Timestamp-derived. The Replica *office0* time additionally includes evaluation/video generation.

^b Mapping time only; excludes the per-view pose refinement used to obtain the reported CaRtGS reconstruction score.

using substantially less image-space supervision. In this run, it also completes mapping in 89.15 s rather than 103.34 s. The workload and fidelity adaptation responsible for this reduction are shown in Fig. 6.

The same map remains available for continued optimization after acquisition. Table II reports separate refinement runs with 4k, 8k, and 16k additional updates. Appearance quality increases from 23.27 dB at the mapping endpoint to 25.62 dB after 16k updates. Representative-view depth RMSE improves mainly within the first 4–8k updates and changes little thereafter.

C. Ablation Study

Table III isolates Load-Adaptive Fidelity and Targeted Geometry Growth across three RGB-D scenes. Endpoint PSNR measures final global quality, while CVQ-AUC and CUC@20 capture regional quality during acquisition. Supervised pixels and Gaussian count measure optimization work and map size. The results should therefore be read as quality–compute trade-offs rather than as a single endpoint ranking.

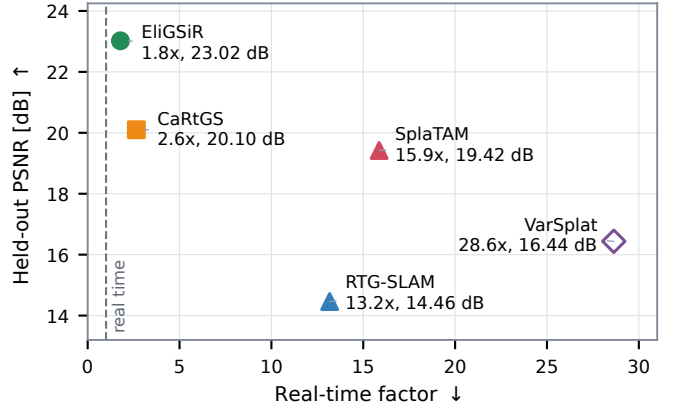


Fig. 3: **Held-out reconstruction quality versus real-time factor on TUM RGB-D fr3/long_office_household.** RTF is elapsed processing time divided by the 87.14 s acquisition duration; 1 $\times$ denotes real-time processing. SplaTAM, RTG-SLAM, and VarSplat use the common ground-truth mapping poses, while CaRtGS uses its native tracker and EliGSiR uses live ORB-SLAM3 poses.

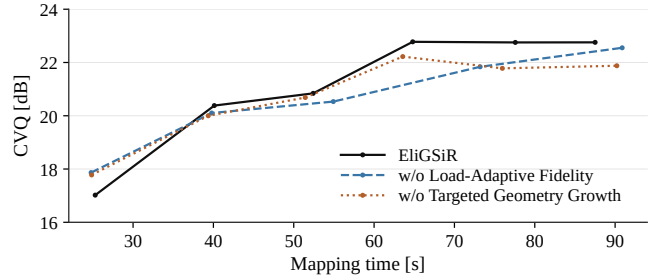


Fig. 4: **Reconstruction quality during mapping on TUM RGB-D fr3/long_office_household.** CVQ over mapping time for EliGSiR, w/o Load-Adaptive Fidelity, and w/o Targeted Geometry Growth. The corresponding source-progress CVQ-AUC values are 21.18, 20.62, and 20.84 dB, respectively.

a) *Load-Adaptive Fidelity*: Figure 6 shows how supervision fidelity follows the current mapping load. On TUM, adaptive fidelity reduces supervision from 2786.9 M to 398.2 M pixels (7.0 $\times$) and mapping time from 103.34 s to 89.15 s, while increasing CVQ-AUC from 20.62 to 21.18 dB and CUC@20 from 65.4% to 69.2%. Fixed native resolution reaches the higher endpoint PSNR (19.36 dB versus 18.50 dB), showing that adaptive fidelity improves online quality for substantially less image-space work rather than maximizing the final score. Supervised pixels also decrease by 1.46 $\times$ on ScanNet++ and 7.9 $\times$ on kitchen1.

b) *Targeted Geometry Growth*: Targeted Geometry Growth adds capacity where repeated RGB-D observations show that the current geometry is insufficient. It increases CUC@20 by 6.1, 5.5, and 2.0 percentage points on TUM, ScanNet++, and kitchen1, respectively. On ScanNet++, it also reduces the representation from 1.109 M to 366.5 k Gaussians while maintaining similar endpoint PSNR (18.71 dB versus 18.59 dB).

TABLE II: **Extended optimization on TUM RGB-D fr3/long_office_household.** Separate 4k, 8k, and 16k refinement runs use a 45-view split distinct from Table I. RGB metrics use the original holdout poses after training-pose alignment. Depth RMSE uses representative retained views.

Map state	Updates	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Depth RMSE [m] $\downarrow$	Gaus. [k]	Total time [s]
ME	1372	23.27	0.807	0.269	0.459	328.1	144.9
+4k PM	5372	23.64	0.814	0.264	0.356	356.6	1166.5
+8k PM	9351	24.44	0.837	0.235	0.347	367.5	2188.6
+16k PM	17102	25.62	0.855	0.213	0.352	381.3	5062.1

TABLE III: **Online ablations across three RGB-D scenes.** Endpoint PSNR measures final global quality; CVQ-AUC and CUC@20 measure regional quality during acquisition. Supervised pixels and Gaussian count report optimization work and map size. Runs are separate from Table I.

Variant	PSNR $\uparrow$	CVQ-AUC $\uparrow$	CUC@20 $\uparrow$	Sup. px. [M] $\downarrow$	Gaus. [k]
TUM RGB-D fr3/long_office_household					
EliGSiR	18.50	21.18	69.2	398.2	140.1
w/o adaptive fidelity	19.36	20.62	65.4	2786.9	549.5
w/o targeted growth	17.78	20.84	63.1	330.9	120.5
ScanNet++ 8b5caf3398					
EliGSiR	18.71	16.28	23.2	17559.6	366.5
w/o adaptive fidelity	18.58	16.36	16.4	25613.1	341.5
w/o targeted growth	18.59	16.59	17.7	14242.3	1108.8
Orbbec kitchen1					
EliGSiR	18.25	19.75	55.3	285.6	123.9
w/o adaptive fidelity	22.91	20.44	77.9	2267.1	565.0
w/o targeted growth	18.63	19.55	53.3	276.9	70.7

c) *Map-Guided View Scheduling*: Map-Guided View Scheduling redirects optimization toward weak regions. Figure 5 shows the fixed world-space microcell grid aligned with the Gaussian map, where each cell summarizes quality for the local mapped surface. Mean and median regional quality remain similar, while several weak regions improve by more than 10 dB. The scheduler therefore redistributes optimization toward regions that still need it rather than uniformly increasing endpoint quality.

D. Qualitative Results

Figures 7 and 8 provide complementary qualitative results. The former shows EliGSiR mapping-end reconstructions on ScanNet and ScanNet++, while the latter compares reference, CaRtGS, and EliGSiR RGB and depth renderings from matched viewpoints on TUM RGB-D and the real-sensor kitchen1 sequence.

VII. DISCUSSION

EliGSiR treats continual RGB-D Gaussian mapping as a compute-allocation problem across view selection, supervision fidelity, and geometry growth. The experiments show that these decisions affect different parts of the compute-quality trade-off: Load-Adaptive Fidelity reduces supervision work, Targeted Geometry Growth controls where representation capacity is added, and Map-Guided View Scheduling redistributes optimization according to regional reconstruction

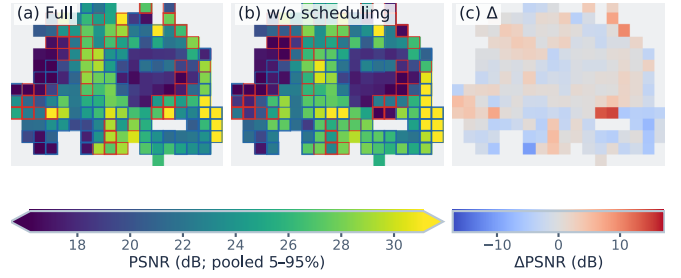


Fig. 5: **Spatial effect of Map-Guided View Scheduling.** Top-down view of the fixed world-space microcell grid aligned with the Gaussian map. Each shown cell summarizes endpoint reconstruction quality for the local mapped surface. We compare the full method, the scheduling ablation, and their difference ($\Delta\text{PSNR} = \text{Full} - \text{w/o scheduling}$) over the same observable cells. Scheduling leaves aggregate quality similar, but redistributes reconstruction quality across the mapped scene.

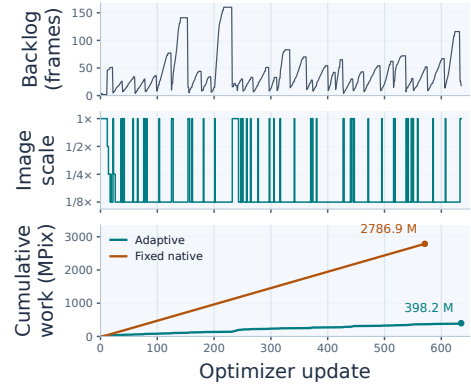


Fig. 6: **Load-Adaptive Fidelity on TUM RGB-D fr3/long_office_household.** Mapping backlog, selected supervision scale, and cumulative image-space supervision over optimization. The controller repeatedly moves between $1/8\times$, $1/4\times$, $1/2\times$, and $1\times$ resolution as mapping load changes, reducing cumulative supervision from 2786.9 M to 398.2 M pixels compared with fixed native resolution.

state. The same Gaussian map remains available for further refinement after acquisition. Real-time operation depends on the sustained workload and currently requires manual tuning of several runtime parameters. Representation size also grows with the explored scene. EliGSiR assumes a predominantly static environment and a sufficiently consistent input trajectory; pose errors, dynamic objects, and unreliable depth can still produce inconsistent geometry. Integrating a visual odometry frontend would extend EliGSiR from mapping to a complete RGB-D SLAM system.

REFERENCES

- [1] B. Kerbl *et al.*, “3D Gaussian Splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.
- [2] N. Keetha *et al.*, “SplaTAM: Splat, track & map 3D Gaussians for dense RGB-D SLAM,” in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition*, 2024, pp. 21357–21366.
- [3] H. Matsuki *et al.*, “Gaussian splatting SLAM,” in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition*, 2024.
- [4] Z. Peng *et al.*, “RTG-SLAM: Real-time 3D reconstruction at scale using Gaussian Splatting,” in *ACM SIGGRAPH Conf. Papers*, 2024.

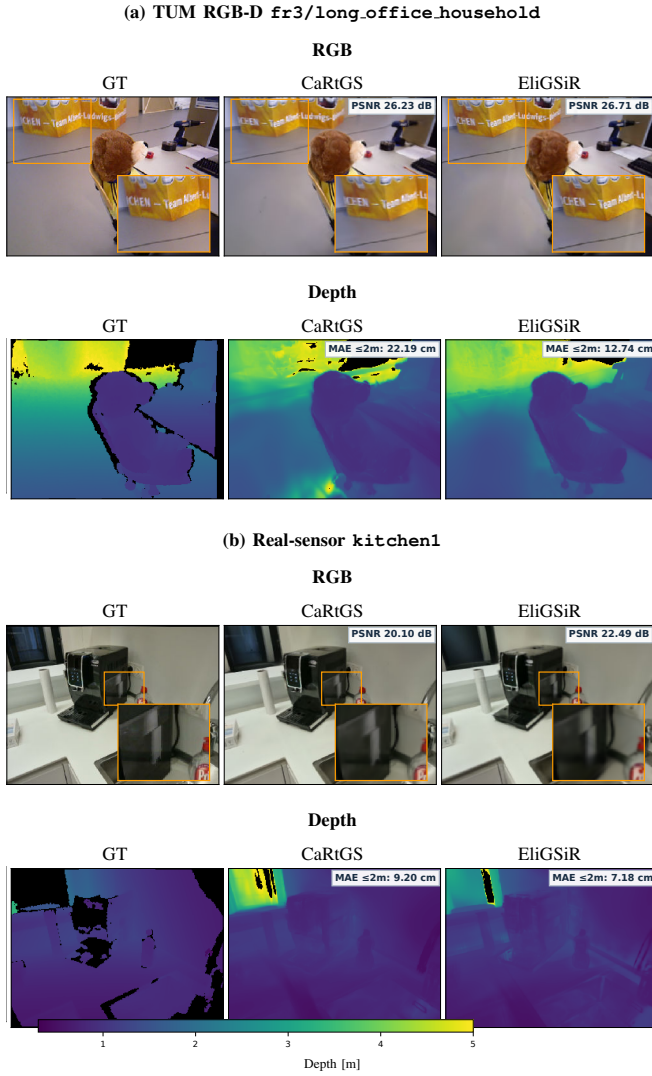


Fig. 8: **Qualitative RGB and depth reconstruction.** Reference, CaRtGS, and EliGSiR renderings from matched viewpoints on TUM RGB-D fr3/long.office.household and the real-sensor kitchen1 sequence.

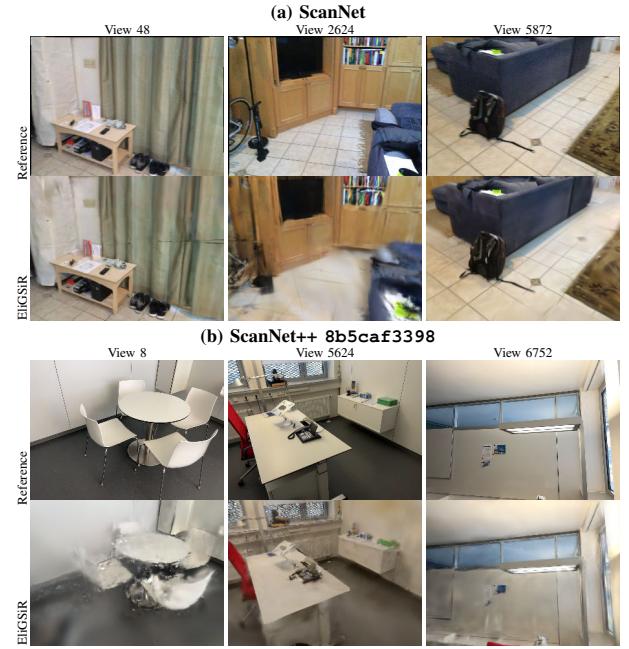


Fig. 7: **Qualitative reconstruction across additional indoor datasets.** Reference images and EliGSiR mapping-end renderings at representative viewpoints on (a) ScanNet and (b) ScanNet++ 8b5caf3398.

- [5] S. Ha *et al.*, “RGBD GS-ICP SLAM,” in *European Conf. on Computer Vision*, 2024, pp. 180–197.
- [6] V. Yugay *et al.*, “Gaussian-SLAM: Photo-realistic dense SLAM with Gaussian Splatting,” *arXiv preprint arXiv:2312.10070*, 2023.
- [7] H. Huang *et al.*, “Photo-SLAM: Real-time simultaneous localization and photorealistic mapping for monocular, stereo, and RGB-D cameras,” in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition*, 2024.
- [8] C. Yan *et al.*, “GS-SLAM: Dense visual SLAM with 3D Gaussian Splatting,” in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition*, 2024.
- [9] L. Zhu *et al.*, “LoopSplat: Loop closure by registering 3D Gaussian Splats,” in *Int. Conf. on 3D Vision*, 2025.
- [10] Z. Zhu *et al.*, “NICE-SLAM: Neural implicit scalable encoding for SLAM,” in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition*, 2022.
- [11] E. Sandström *et al.*, “Point-SLAM: Dense neural point cloud-based SLAM,” in *IEEE/CVF Int. Conf. on Computer Vision*, 2023.
- [12] D. Feng *et al.*, “CaRtGS: Computational alignment for real-time Gaussian Splatting SLAM,” *IEEE Robotics and Automation Letters*, 2025.
- [13] J. Wei *et al.*, “GSFusion: Online RGB-D mapping where Gaussian Splatting meets TSDF fusion,” *IEEE Robotics and Automation Letters*, 2024.
- [14] S. Kheradmand *et al.*, “3D Gaussian Splatting as Markov Chain Monte Carlo,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024.
- [15] J. Straub *et al.*, “The Replica dataset: A digital replica of indoor spaces,” *arXiv preprint arXiv:1906.05797*, 2019.
- [16] J. Sturm *et al.*, “A benchmark for the evaluation of RGB-D SLAM systems,” in *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems*, 2012, pp. 573–580.
- [17] A. Dai *et al.*, “ScanNet: Richly-annotated 3d reconstructions of indoor scenes,” in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition*, 2017.
- [18] C. Yeshwanth *et al.*, “ScanNet++: A high-fidelity dataset of 3D indoor scenes,” in *IEEE/CVF Int. Conf. on Computer Vision*, 2023.
- [19] V. Ye *et al.*, “gsplat: An open-source library for Gaussian Splatting,” *Journal of Machine Learning Research*, vol. 26, no. 34, pp. 1–17, 2025.
- [20] C. Campos *et al.*, “ORB-SLAM3: An accurate open-source library for visual, visual-inertial and multi-map SLAM,” *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.